



Disentangling Hype from Reality for Artificial Intelligence–Based Skin Cancer Diagnosis: Comment on a Narrative Review

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The explosion of claims around the performance of artificial intelligence (AI) for dermatology clouds the ability to separate the hype from the reality. In “Artificial intelligence in skin cancer diagnosis: a reality check,” Brancaccio et al (2023) review the state of image-based AI diagnostics in dermatology, with a particular focus on real-world impact for consumers, general practitioners, and dermatologists. Their review reveals that although there are plenty of opportunities and excitement around image-based AI for dermatology, there remains a significant need for additional trials and evaluation in real-world settings.

Despite the need for additional validation of image-based AI, numerous direct-to-consumer (DTC) applications have already been released (Brancaccio et al, 2023). A survey conducted in the United Kingdom found that 41% of people would trust an AI-based application to diagnose skin cancers (British Association of Dermatologists, 2022). However, Brancaccio et al (2023) highlight prior work showing that the accuracy of these algorithms is variable on an independent test set of clinical images: of 25 apps available in 2022, mean diagnostic accuracy was between 50 and 60% (Brancaccio et al,

2023; Sun et al, 2022). Of note, newer consumer algorithms tested in early prospective trials in Austria showed some promise (sensitivity/specificity > 90%). However, such evidence is still preliminary and used physician-captured images rather than patient-captured images (Kränke et al, 2023). Much of existing research on DTC AI applications is plagued by poor methodological quality, including high risk of bias, such as evaluation of high-quality images captured by clinicians as opposed to real-world patient images, and poor generalizability attributable to evaluation under artificial conditions, such as assessment of diagnostic accuracy in isolation without additional patient medical history (Brancaccio et al, 2023). Although transparency with regard to study design has improved after the publication of the SPIRIT (Standard Protocol Items: Recommendations for Interventional Trials) and CONSORT-AI (Consolidated Standards of Reporting Trials-Artificial Intelligence) guidelines (Liu et al, 2020a; Rivera et al, 2020), AI developers are not required to list evidence supporting the diagnostic capabilities of their algorithms in the DTC interface, raising concerns about consumer deception, especially when important diagnostic information is delivered in the absence of adequate counseling by medical professionals (British Association of Dermatologists, 2022).

Image-based AI diagnostics show some promise in aiding general practitioners and potentially improving access by enabling telemedicine. Brancaccio et al (2023) describe 2 previous studies in which nondermatologists reported benefits/improvement in diagnostic accuracy with AI assistance. However, in general, algorithms perform poorly on images underrepresented in their training datasets, which tend to consist of common conditions on fair skin in idealized settings rather than the full clinical spectrum of disease on diverse skin tones (Daneshjou et al, 2022).

Research for augmenting dermatologist performance is more plentiful but indicates that we are still in the early days of deciphering what tools will actually provide meaningful clinical impact. Studies evaluating AI against dermatologists have been conducted in artificial settings that are not reflective of real-world practice. In fact, when AI models are pitted against dermatologists in a real-world setting, the dermatologists outperform the models (Brancaccio et al, 2023). The studies that do show promise have focused on dermatologist–AI collaboration, with

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Clinical Implications

- Lack of transparency, biased training datasets, and evaluation under artificial conditions limit the strength of current evidence for image-based artificial intelligence (AI) diagnostics.
- Direct-to-consumer applications have the least robust evidence, whereas dermatologists may not uniformly benefit from AI assistance.
- Other types of foundation models may be useful for improving explainability, training data, and accuracy.

algorithms proving to be most useful for novice dermatologists (Brancaccio et al, 2023). However, what is needed now is additional prospective trials where dermatologist–AI collaboration is tested in real-world clinical settings to ensure patient benefit.

As we look toward the future, although standalone image-based diagnostic algorithms have their advantages, the field is increasingly moving toward algorithms that more closely parallel the working model of the clinician, who (i) synthesizes multiple sources of information and (ii) references trusted text resources to arrive at a final diagnosis and treatment plan. The first half of this analogy can be done by multimodal models, which can, in theory, incorporate multiple inputs and input types, such as images, medical history, laboratory results, pathology interpretation, genomics, and time course, into the final diagnosis. Early examples of multimodal models include work by Liu et al (2020b), whose model included images and clinical information such as patient demographics and clinical history. The second half of the analogy can be seen with large language models that use retrieval-augmented generation (RAG), which combines the generative power of large language models with non-generative retrieval models that serve as an open-book source of knowledge. Besides enabling a higher degree of accuracy by conditioning outputs on high-quality sources (rather than the entire Internet), RAGs (such as Clinfo.ai, which references PubMed in a manner roughly analogous to a clinician who consults PubMed or UpToDate) can also provide the end user with the passages the model used to derive its answer, increasing

transparency and simplifying the verification process (Lozano et al, 2023).

In summary, although the field of image-based diagnostics is rapidly evolving, current research is insufficient to suggest definite clinical utility. Perhaps the most evidence exists for dermatologist-focused augmented intelligence applications, although even within this indication, more robust studies will be needed to adequately assess real-world value and safety. Future studies should focus on increasing transparency and real-world validation (in terms of conditions seen, diverse skin tones, and realistic study workflows). With the rise of large language models and multimodal large-scale models, the future of AI in dermatology will also likely include models that incorporate multiple sources of information, more closely mimicking clinician workflows.

CONFLICT OF INTEREST

RD has served as an advisor to MDAI Algorithms and Revea and received consulting fees from Pfizer, L'Oreal, Frazier Healthcare Partners, and DWA and research funding from UCB. The remaining author states no conflict of interest.

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